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MOS Transistor Theory

1 Introduction to MOS Transistor

- A MOS transistor is a **majority carrier device**, in which the current in the channel between the **source** and **drain** is modulated by a voltage applied to the **gate**.

1.1 nMOS Transistor

- Majority carriers are **electrons**.
- A **positive** gate voltage relative to the substrate increases electron concentration in the channel, enhancing conductivity.
- For **gate voltages below the threshold voltage** V_t, the channel is cut off, leading to a very low drain-to-source current.

1.2 pMOS Transistor

- Majority carriers are **holes**.
- A **negative** gate voltage enhances conduction.
- The operation is analogous to nMOS but with opposite voltage polarities.

1.3 Threshold Voltage (V_t)

- The **threshold voltage** V_t is the voltage at which a MOSFET begins to conduct.
- The source-to-drain current I_{ds} depends on the gate-to-source voltage V_{gs}.

1.4 Enhancement vs. Depletion Mode

There are two types of MOS transistors based on conduction behavior at zero gate bias:

1. **Enhancement Mode:** The transistor is **normally OFF** (requires a gate voltage to conduct).
2. **Depletion Mode:** The transistor is **normally ON** (conducts even at zero gate voltage).

1.5 nMOS vs. pMOS Transistors

- nMOS and pMOS transistors are **duals of each other**, meaning they function similarly but with opposite voltage polarities.

- The threshold voltages are denoted as:
  V_{tn} : (n-channel MOSFET)
  V_{tp} : (p-channel MOSFET)

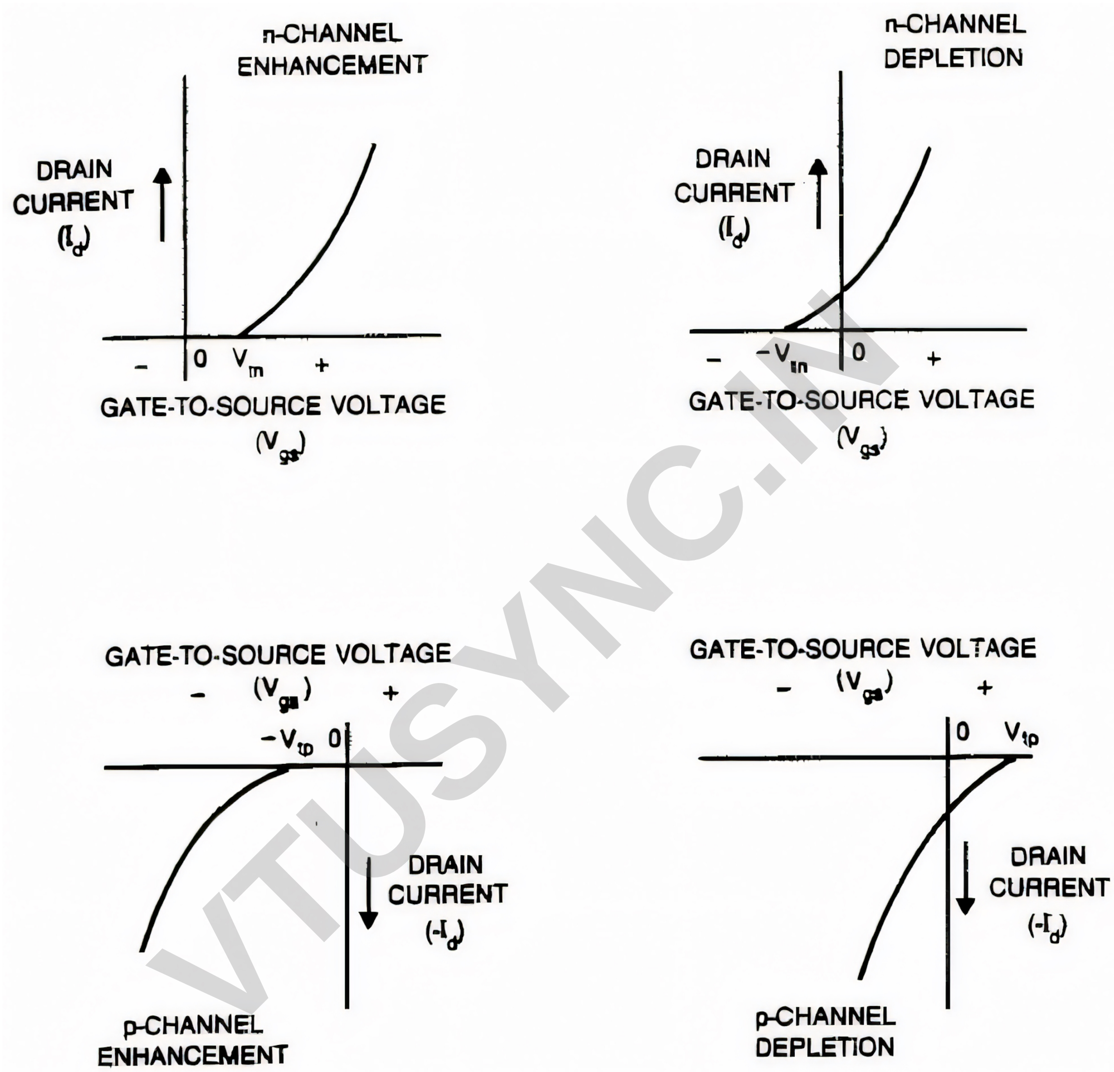

Figure 1: Conduction characteristics for enhancement and depletion mode transistors (assuming fixed V_{ds})

1.6 CMOS Technology

- Both nMOS and pMOS transistors are used together in **CMOS (Complementary MOS) circuits**.
- Most modern CMOS integrated circuits use **enhancement-mode transistors**.

2 n-MOS Enhancement Transistor

- An nMOS enhancement-mode transistor is a voltage-controlled device widely used in digital and analog circuits.
- It operates based on an electric field-induced conduction channel between its source and drain terminals.
- Unlike bipolar junction transistors (BJTs), where charge carriers are introduced by doping, an nMOS transistor forms a conductive channel through an applied gate voltage.

2.1 Structure of nMOS Enhancement-Mode Transistor

- The basic structure of an nMOS enhancement-mode transistor is shown in figure below. It consists of:
  - A **moderately doped p-type silicon substrate**.
  - Two **heavily doped n-type regions** called the source and drain, diffused into the substrate.
  - A **thin insulating layer** of silicon dioxide (SiO_2) covering the channel region, known as the **gate oxide**.
  - A **polycrystalline silicon (polysilicon)** gate electrode over the oxide layer.
- Since SiO_2 is an excellent insulator, the gate does not allow direct current flow to the channel.

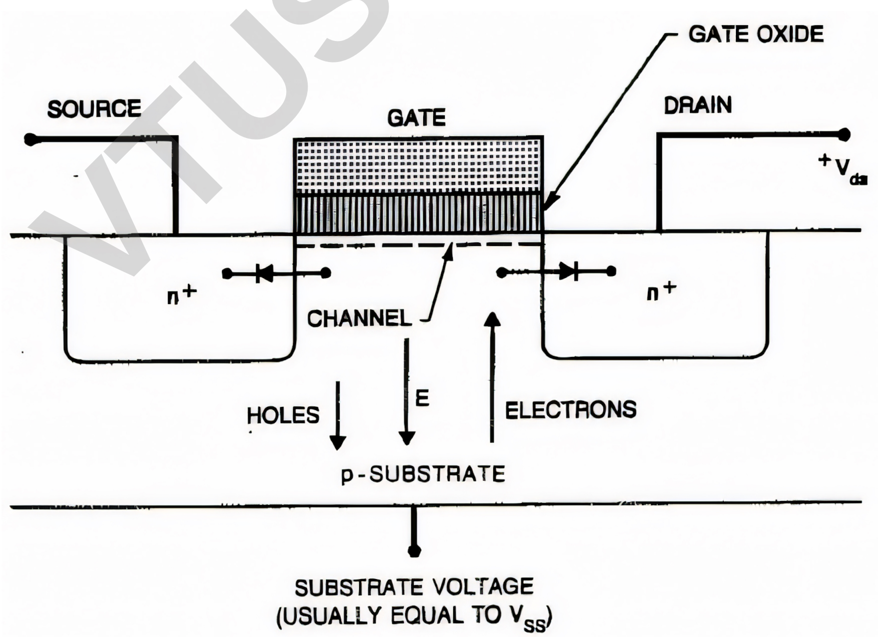

Figure 2: Physical structure of an nMOS transistor

- The transistor is symmetrical, meaning the source and drain can be interchanged physically.
- However, in circuit design, the source is usually defined as the terminal at a lower potential.

2.2 Principle of Operation

The operation of the nMOS transistor depends on the applied gate-source voltage (V_{GS}) and drain-source voltage (V_{DS}).

2.2.1 With Zero Gate Bias ($V_{GS} = 0$)

- The p-type substrate prevents current flow between the source and drain because the two p-n junctions (between the n+ regions and the p-substrate) are reverse biased.
- The source and drain act as two isolated n-regions, with only leakage current flowing.

2.2.2 Applying a Positive Gate Voltage ($V_{GS} > 0$)

- The positive voltage on the gate creates an electric field (E) that repels holes and attracts electrons toward the oxide-silicon interface.
- When V_{GS} exceeds a threshold voltage (V_t), sufficient electrons accumulate to form an **inversion layer**, effectively turning the p-type region beneath the gate into an n-type channel.
- This allows current to flow between source and drain if a voltage V_{DS} is applied.

2.2.3 Field-Induced vs. Metallurgical Junction

- Unlike BJTs, where n-type conductivity is introduced via doping, the MOSFET channel is induced by an electric field.
- This field-induced junction enables voltage-controlled operation.

2.3 Operating Regions of nMOS Transistor

The nMOS transistor operates in three regions based on the applied voltages:

1. **Cutoff Region ($\mathbf{V_{GS} < V_t}$):**
   - The gate voltage is below the threshold voltage.
   - No conduction occurs except for a small leakage current.
   - The transistor acts as an open switch.
2. **Linear (Triode) Region ($\mathbf{V_{GS} > V_t}$ and $\mathbf{V_{DS} < V_{GS} - V_t}$):**
   - The inversion layer forms a conductive channel between the source and drain.
   - The transistor behaves like a voltage-controlled resistor.
   - Drain current I_{DS} is approximately proportional to V_{DS}.
   - The drain current in this region is given by:

$$I_{DS} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{V_{DS}^2}{2} \right] \tag{1}$$

   where:

μ_n is the electron mobility.
C_{ox} is the gate oxide capacitance per unit area.
W is the channel width.
L is the channel length.

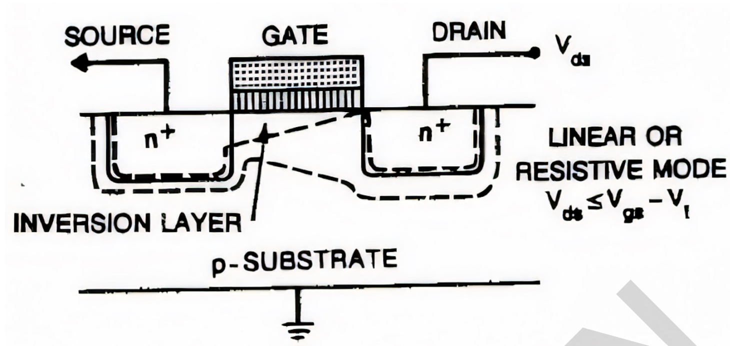

Figure 3: nMOS in Linear region

3. **Saturation Region ($\mathbf{V_{GS} > V_t}$ and $\mathbf{V_{DS} > V_{GS} - V_t}$):**
   - The channel becomes **pinched off** at the drain end.
   - The drain current is primarily controlled by V_{GS} and is almost independent of V_{DS}.
   - The drain current in this region is given by:

$$I_{DS} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_t)^2 \tag{2}$$

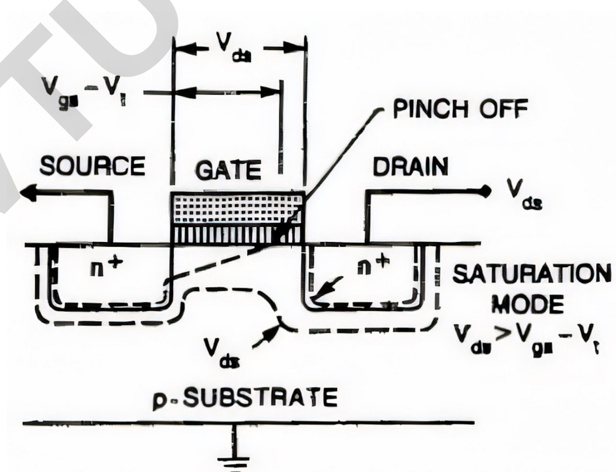

Figure 4: nMOS in Saturation region

- **Pinch-Off and Channel Behavior**
  - At the source end, the full gate voltage is effective.
  - At the drain end, only $(V_{GS} - V_{DS})$ is effective.
  - When $V_{DS} > V_{GS} - V_t$, the channel is **pinched off**.
  - Electrons drift toward the drain due to the electric field.

2.3.1 Factors Affecting Drain Current

The drain current I_{DS} is influenced by:

- The distance between source and drain, i.e., channel length (L).
- The channel width (W).
- The threshold voltage (V_t).
- The thickness of the gate-insulating oxide layer.
- The dielectric constant of the gate insulator.
- Carrier mobility (μ_n).

2.3.2 Breakdown and Abnormal Conditions

1. Avalanche Breakdown

- If a very high voltage is applied to the drain, impact ionization occurs.
- This leads to excessive current flow, causing possible device failure.

2. Punch-Through Effect

- If V_{DS} is too high, the depletion region extends from the source to drain, allowing direct current flow.
- This results in loss of gate control over the drain current.

2.3.3 Summary

Region	Gate Voltage (V_{GS})	Drain Voltage (V_{DS})	Current Flow (I_{DS})
Cutoff	$V_{GS} < V_t$	Any	No conduction (only leakage current)
Linear	$V_{GS} > V_t$	$V_{DS} < V_{GS} - V_t$	Increases linearly with V_{DS}
Saturation	$V_{GS} > V_t$	$V_{DS} > V_{GS} - V_t$	Controlled by V_{GS}, nearly independent of V_{DS}

2.4 Conclusion

- The nMOS enhancement-mode transistor is an essential component in digital and analog circuits.
- It operates as a voltage-controlled switch, allowing conduction only when $V_{GS} > V_t$.
- By varying the gate and drain voltages, the device transitions between different operating regions, making it ideal for digital logic and analog applications.
- Understanding its different operating regions and the impact of design parameters is crucial for efficient circuit design.

3 pMOS Transistor

- While most discussions on MOS transistors focus on **nMOS** devices, a complementary type called the **pMOS** transistor exists.
- A pMOS transistor is structurally similar to an nMOS transistor but with reversed doping polarities.
- This means the substrate is n-type, and the source and drain are p-type.
- The pMOS transistor operates in a manner similar to an nMOS transistor but with opposite voltage polarities and charge carrier movement.

3.1 Structure of a pMOS Enhancement-Mode Transistor

A pMOS enhancement-mode transistor consists of:

- An **n-type silicon substrate**.
- Two heavily doped **p-type regions**, called the source and drain, diffused into the substrate.
- A **thin insulating layer** of silicon dioxide (SiO_2) covering the channel region, known as the **gate oxide**.
- A **polycrystalline silicon (polysilicon)** gate electrode over the oxide layer.

Figure below illustrates the structure of a pMOS transistor.

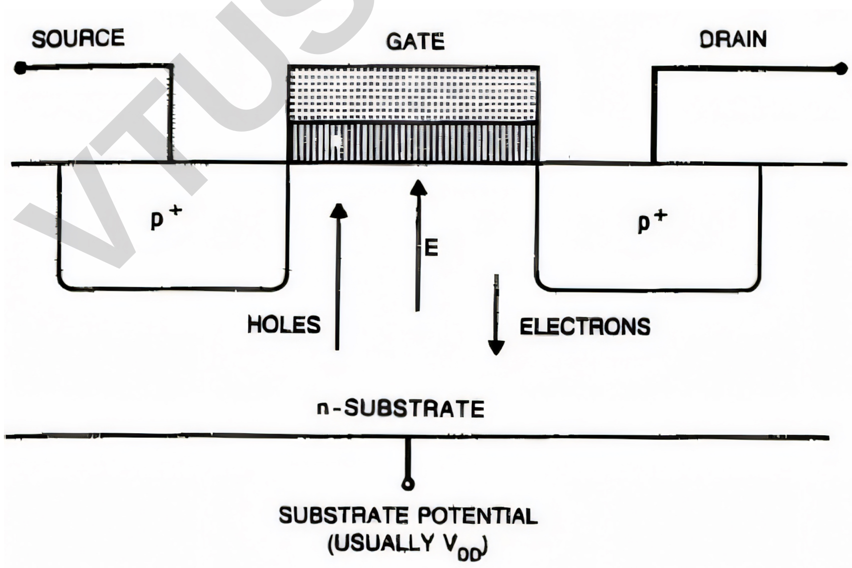

Figure 5: Physical structure of PMOS transistor

3.2 Operation of a pMOS Transistor

Similar to the nMOS transistor, a pMOS transistor operates based on the creation of a conductive channel between the source and drain when a proper gate voltage is applied.

3.2.1 With Zero Gate Bias ($V_{GS} = 0$)

- No conduction occurs because the source and drain are separated by two reverse-biased p-n junctions.
- The source and drain act as two isolated p-regions in an n-type substrate.

3.2.2 Applying a Negative Gate Voltage ($V_{GS} < 0$)

- A negative voltage on the gate repels electrons and attracts holes toward the SiO_2-silicon interface.
- When V_{GS} is sufficiently negative (i.e., less than a threshold voltage V_t), an **inversion layer** forms, converting the n-type region under the gate into a p-type conductive channel.
- This allows holes to flow from the source to the drain if a negative drain-to-source voltage (V_{DS}) is applied.

3.2.3 Charge Carrier Movement

- In an **nMOS** transistor, conduction occurs due to electron movement.
- In a **pMOS** transistor, conduction occurs due to **hole movement**.
- A **negative drain voltage** causes holes to move from the source through the channel to the drain.

3.3 Comparison Between pMOS and nMOS Transistors

Parameter	nMOS	pMOS
Substrate Type	p-type	n-type
Source/Drain Doping	n+ regions	p+ regions
Charge Carriers	Electrons	Holes
Gate Control	Positive voltage creates n-channel	Negative voltage creates p-channel
Drain Voltage	Positive for conduction	Negative for conduction
Carrier Mobility	High	Lower than electrons

3.4 Conclusion

- The pMOS transistor operates in a manner similar to an nMOS transistor but with reversed voltage polarities and charge carrier movement.
- While nMOS transistors dominate modern digital circuits due to higher electron mobility, pMOS transistors are still used in complementary MOS (CMOS) technology to achieve low power consumption and enhanced performance.

4 Threshold Voltage (V_t)

- The **threshold voltage**, V_t, of an MOS transistor is the minimum gate-to-source voltage (V_{GS}) required to create a conductive inversion layer (channel) between the source and drain terminals.
- Below this voltage, the drain-to-source current (I_{DS}) is negligible, meaning the transistor is in the *cut-off* region.

4.1 Factors Affecting Threshold Voltage

The threshold voltage is influenced by several factors, including:

- **Gate material**: Different materials have different work functions, which affect the required voltage for inversion.
- **Gate insulation material**: The dielectric constant of the insulator affects the threshold voltage.
- **Gate insulator thickness**: Thinner oxide layers result in stronger electric fields and lower threshold voltages.
- **Channel doping**: Heavily doped substrates increase the threshold voltage.
- **Impurities at the silicon-insulator interface**: Trapped charges and interface states can shift the threshold voltage.
- **Voltage between source and substrate (V_{SB})**: Also known as the **body effect**, an increase in V_{SB} raises the threshold voltage.

4.2 Temperature Dependence of Threshold Voltage

The absolute value of the threshold voltage decreases with increasing temperature. This variation is approximately:

- -4 mV/°C for high substrate doping levels.
- -2 mV/°C for low substrate doping levels.

As temperature rises, increased carrier generation reduces the depletion region width, leading to a lower threshold voltage.

4.3 Conclusion

- The threshold voltage is a key parameter in MOS transistor operation, determining when the device transitions from the off-state to conduction.
- It depends on material properties, doping levels, and external bias conditions, and exhibits temperature sensitivity that must be considered in circuit design.

5 Threshold Voltage Adjustment

- The native (original) threshold voltage (V_t) of an MOS transistor may not always be suitable for a given application.
- Therefore, it is often necessary to adjust V_t to achieve desired device characteristics.
- Two common techniques used for threshold voltage adjustment are:
  1. **Varying the doping concentration at the silicon-insulator interface through ion implantation**.
  2. **Using different insulating materials for the gate dielectric**.

5.1 Ion Implantation for Threshold Voltage Control

- One effective method to modify the threshold voltage is through **ion implantation**, which alters the doping concentration at the silicon-insulator interface.
- This technique allows precise control over V_t by introducing dopant atoms into the channel region.

5.2 Using Different Gate Insulating Materials

- Another approach to adjusting the threshold voltage is by modifying the gate dielectric material.
- Instead of using only silicon dioxide (SiO_2), a combination of silicon nitride (Si_3N_4) and silicon dioxide can be used.
  - **Silicon nitride (Si_3N_4)** has a relative permittivity of 7.5.
  - **Silicon dioxide (SiO_2)** has a relative permittivity of 3.9.
  - When a layer of silicon nitride is combined with a layer of silicon dioxide, the resulting **effective permittivity** is about 6.
- Since the effective permittivity of the dual-dielectric structure is higher than that of SiO_2 alone, the dual-layer dielectric is electrically equivalent to a thinner SiO_2 layer of the same physical thickness.
- This reduces the required gate voltage to achieve inversion, effectively lowering V_t.

5.3 Preventing Unwanted Inversion

To ensure that the silicon surface does not invert in regions between transistors, the threshold voltage in these **field regions** is increased by:

- Heavily doped diffusions.
- Ion implantation into the silicon surface.
- Increasing the oxide layer thickness in these regions.

5.4 Self-Isolation of MOS Transistors

MOS transistors are inherently **self-isolating** as long as:

- The silicon surface inverts under the gate.
- The surface does not invert in the regions between transistors under normal circuit voltage conditions.

5.5 Conclusion

- Threshold voltage adjustment is an essential technique in MOS transistor design, allowing fine-tuning of device performance.
- Ion implantation and high-permittivity gate dielectrics provide effective means to control V_t, ensuring proper operation and isolation between transistors in integrated circuits.

6 Body Effect

- In MOSFETs, the **body effect** (also known as the **substrate bias effect**) occurs when the voltage difference between the **source** and **substrate** (V_{sb}) affects the **threshold voltage** (V_t).
- This effect becomes significant in circuits where multiple transistors are connected in series, such as in CMOS logic gates, because the substrate voltage is usually common for all transistors.

6.1 Common Substrate and its Impact

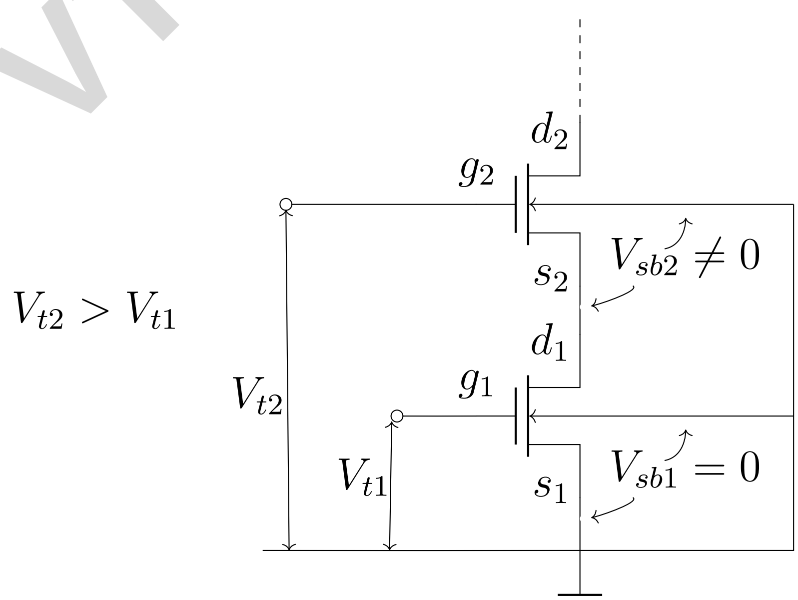

Figure 6: The effect of body effect on series connected transistors

- In most MOS circuits, all transistors share a **common substrate**.
- Under normal conditions, the substrate voltage remains the same for all transistors.
- However, in certain configurations — especially when multiple MOSFETs are connected in series — this assumption no longer holds.
- Consider a chain of MOSFETs connected in series, as shown in figure above.
- The first transistor in the series may have its source directly connected to ground, leading to a source-to-substrate voltage of **zero** ($V_{sb1} = 0$).
- However, for the second transistor in the series, its source is connected to the drain of the first transistor, which is at a **higher potential**.
- This results in a nonzero source-to-substrate voltage ($V_{sb2} \neq 0$).
- As we move along the series chain, V_{sb} continues to increase.

6.2 How the Body Effect Alters the Threshold Voltage

- The **threshold voltage** of a MOSFET is the minimum gate-to-source voltage (V_{gs}) required to form a conductive channel between the source and drain.
- Under normal conditions, when $V_{gs} > V_t$, a **channel** is formed, and charge carriers (electrons in nMOS or holes in pMOS) flow from the source to the drain.
- When V_{sb} increases, it affects the **depletion region** at the substrate-channel junction:
  - The **depletion layer width** increases.
  - More **charge carriers** get trapped in the depletion layer.
  - To maintain **charge neutrality**, the available **channel charge decreases**.
- Since the channel charge is reduced, the gate must apply **more voltage** to invert the channel.
- This means that the **threshold voltage** (V_t) increases, making it harder to turn the transistor **ON**.
- This phenomenon is called the **body effect**.

6.3 Mathematical Expression for the Body Effect

The **modified threshold voltage** in the presence of a nonzero V_{sb} can be approximated using the following equation:

$$V_t = V_t(0) \pm \gamma\sqrt{V_{sb}} \tag{3}$$

where:

- $V_t(0)$ is the threshold voltage when $V_{sb} = 0$ (i.e., when the source and substrate are at the same potential).

- γ is the **body-effect coefficient**, which depends on substrate doping. Typical values range from **0.4 to 1.2**.
- V_{sb} is the **source-to-substrate voltage**.
- The **negative sign** is used for **pMOS** transistors because their substrate is typically at a higher potential than the source.

This equation shows that as V_{sb} increases, the threshold voltage V_t also increases, making the MOSFET less conductive.

6.4 Impact of the Body Effect on Circuit Performance

The increase in V_t due to the body effect has several consequences on circuit behavior:

- Reduced Drain Current (I_D)
- Slower Switching Speeds
- Design Considerations for CMOS Technology

6.5 Conclusion

- The **body effect** plays a crucial role in MOSFET operation, particularly in **series-connected transistors** where V_{sb} is nonzero.
- It leads to an **increase in the threshold voltage**, which in turn reduces the drain current and slows down circuit operation.
- Understanding and managing this effect is essential in **CMOS circuit design**, especially in high-speed and low-power applications.

7 MOS Device Design Equations

MOS transistors operate in three distinct regions, each governed by specific design equations:

1. Cut-off Region
2. Linear (Triode) Region
3. Saturation (Active) Region

Before going into the equations, let us establish few important concepts.

7.1 Important Concepts

7.1.1 Geometric terms:

- Consider the diagram given below.
- The channel length is represented by L.
- The channel width is represented by W.

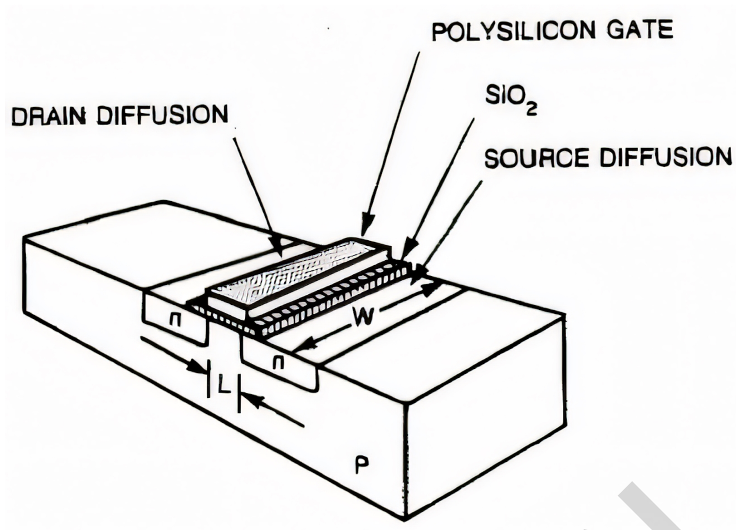

Figure 7: Geometric terms in the MOS device equations

7.1.2 MOS Gain Factor (β)

The MOS gain factor is defined as:

$$\beta = \frac{\mu C_{ox} W}{L} \tag{4}$$

where:

- μ = Effective carrier mobility
- $C_{ox} = \dfrac{\varepsilon_{ox}}{t_{ox}}$ = Gate oxide capacitance per unit area
- t_{ox} = Gate oxide thickness
- W = Channel width
- L = Channel length

A typical calculation for an nMOS device:

$$\beta = \frac{500 \times (4 \times 8.85 \times 10^{-14})W}{5 \times 10^{-5} L} \approx 35 \mu A/V^2$$

7.1.3 Process Gain Factor (K_p)

- $K_p = \dfrac{\mu \epsilon_{ox}}{t_{ox}}$ is known as the **process gain factor**.
- It is a key **SPICE model parameter**.
- Typical values: 10 to 30 $\mu A/V^2$, with a variation of $10 - 20\%$.

7.1.4 The carrier mobility (μ)

- The mobility, μ, describes the ease with which carriers drift in the substrate material.

- It is defined by:

$$\mu = \frac{\text{average carrier drift velocity}(v)}{\text{Electric Field (E)}} \tag{5}$$

- If the velocity (v) is given in cm/sec. and the electric field (E) in V/cm, the mobility has dimensions of $cm^2/V - sec$.

7.1.5 Threshold Voltage and Body Effect

A more accurate expression for the threshold voltage is:

$$V_t = V_{t0} + \gamma \left(\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F} \right) \tag{6}$$

where:

- V_{t0} = Threshold voltage at $V_{SB} = 0$
- γ = Body-effect coefficient
- V_{SB} = Source-to-body voltage
- ϕ_F = Fermi potential

The body-effect coefficient is given by:

$$\gamma = \frac{t_{ox}}{\epsilon_{ox}} \sqrt{2q\epsilon_{Si} N} \tag{7}$$

where:

- q = Charge of an electron
- N = Concentration density of the substrate
- C_{ox} = Gate oxide capacitance
- ε_{si} = Dielectric constant of silicon substrate

7.1.6 Short-Channel Effects and Channel Length Modulation

- The **effective channel length** is:

$$L_{eff} = L - \Delta L \tag{8}$$

  where ΔL accounts for channel reduction at high V_{DS}.
- This reduces **output resistance** and increases **drain conductance**.

7.2 Regions of MOS Transistor Operation

7.2.1 Cut-off Region ($V_{GS} < V_t$)

- The MOS transistor is **off**, and there is no conduction between drain and source ($I_{DS} \approx 0$).
- A small leakage current exists.

7.2.2 Linear Region ($0 < V_{DS} < V_{GS} - V_t$)

- The MOSFET behaves like a **voltage-controlled resistor**.
- The drain current is given by:

$$I_{DS} = \beta\left(V_{GS} - V_t\right)V_{DS} - \frac{1}{2}\beta V_{DS}^2 \tag{9}$$

7.2.3 Saturation Region ($V_{DS} > V_{GS} - V_t$)

- The MOSFET is **fully on** and behaves like a **constant current source**.
- The drain current is:

$$I_{DS} = \frac{1}{2}\beta\left(V_{GS} - V_t\right)^2 \tag{10}$$

- The above equation assumes that the current in the channel saturates (i.e., is constant) and is independent of the applied drain voltage.
- In practice, the drain current in saturation increases slightly with increasing drain voltage. This phenomenon is called the **channel length modulation**
- A more accurate model accounting for **channel length modulation** is:

$$I_{DS} = \frac{1}{2}\beta\left(V_{GS} - V_t\right)^2\left(1 + \lambda V_{DS}\right) \tag{11}$$

7.3 Application to PMOS Devices

- The same equations apply to PMOS transistors.
- The only difference is the **signs** of voltages and currents:
  - For nMOS: V_{GS}, V_{DS}, I_{DS} are positive.
  - For pMOS: V_{GS}, V_{DS}, I_{DS} are negative.

7.4 Summary

- MOS transistors operate in **cut-off, linear, and saturation** regions.
- The **MOS gain factor** depends on mobility, oxide capacitance, and device geometry.
- **Body effect** influences V_t, which must be considered in design.
- **Channel length modulation** affects I_{DS} in saturation.
- PMOS equations are the same as nMOS but with reversed signs.

8 V-I Characteristics

- The voltage-current (V-I) characteristics of **n-channel** and **p-channel** MOSFETs in both the **linear** and **saturation regions** describe how the drain current I_{DS} varies with the applied drain-to-source voltage V_{DS} and gate-to-source voltage V_{GS}.

- In figure below, the characteristics are plotted using the **absolute values** of the voltages so that both **n-type** and **p-type transistors** can be represented on the same axes.

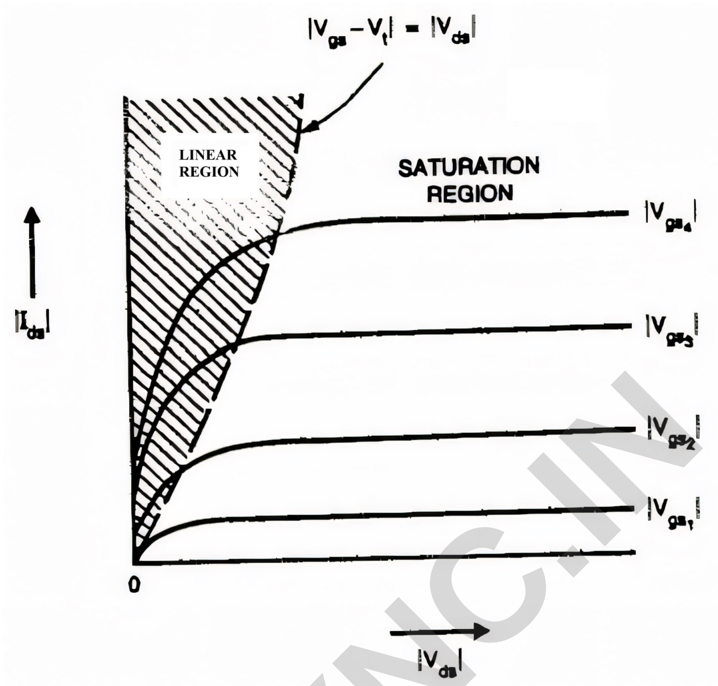

Figure 8: V-1 characterlstics for n- and p-transistors

- The **boundary** between the **linear** and **saturation** regions is determined by the condition:

$$V_{DS} = V_{GS} - V_t \tag{12}$$

where V_t is the **threshold voltage**.

- This boundary appears as a **dashed line** in the characteristic curves.

8.1 Output Resistance in the Linear Region

- In the **linear region**, the MOSFET behaves as a voltage-controlled resistor.
- The **output resistance** (also known as **channel resistance** R_c) can be derived by differentiating the **drain current equation** with respect to V_{DS}.
- The resulting **output conductance** is given by:

$$\lim \frac{dI_{DS}}{dV_{DS}} = \beta(V_{GS} - V_t) \tag{13}$$

- Rearranging the above equation, the **channel resistance** R_c is approximated as:

$$R_c = \frac{1}{\beta(V_{GS} - V_t)} \tag{14}$$

This equation shows that the resistance of the MOSFET channel is controlled by the **gate-to-source voltage** V_{GS}.

- **Note:** The above relation is valid as long as **carrier mobility remains constant** in the channel.

8.2 MOSFET Behavior in the Saturation Region

- In the **saturation region** ($V_{DS} \geq V_{GS} - V_t$), the MOSFET behaves like a **constant current source** rather than a variable resistor.
- From the **MOSFET current equation in saturation**:

$$I_{DS} = \frac{1}{2}\beta(V_{GS} - V_t)^2 \tag{15}$$

it can be seen that **drain current** I_{DS} is nearly independent of V_{DS}.

- This can be verified by differentiating I_{DS} with respect to V_{DS}:

$$\frac{dI_{DS}}{dV_{DS}} \approx 0 \tag{16}$$

- Since the derivative is nearly zero, the drain current remains **constant** despite changes in V_{DS}.
- This is a key property of MOSFETs used as **current sources** in analog circuits.

8.3 Transconductance (g_m)

- Transconductance (g_m) is a key parameter that defines the **gain** of a MOSFET.
- It expresses the relationship between the **output current** (I_{DS}) and the **input voltage** (V_{GS}):

$$g_m = \left.\frac{dI_{DS}}{dV_{GS}}\right|_{V_{DS}=\text{constant}} \tag{17}$$

This parameter determines how efficiently the MOSFET converts a voltage signal at the gate into a current signal at the drain.

8.3.1 Transconductance in the Linear and Saturation Regions

1. **Linear Region:**

$$g_m = \beta V_{DS} \tag{18}$$

This equation shows that in the **linear region**, transconductance is directly proportional to V_{DS}.

2. **Saturation Region:**

$$g_m = \beta(V_{GS} - V_t) \tag{19}$$

In the **saturation region**, transconductance depends only on ($V_{GS} - V_t$), meaning the MOSFET behaves like an **amplifier** with a gain controlled by V_{GS}.

For a **p-type transistor**, absolute values are used for voltages to ensure that **transconductance remains positive**.

8.4 Key Takeaways

- **MOSFETs have distinct operating regions:** Linear (resistive) and Saturation (current source).
- **The boundary between these regions is at** $V_{DS} = V_{GS} - V_t$.
- **Channel resistance** R_c is controlled by V_{GS} and is inversely proportional to $(V_{GS} - V_t)$.
- In **saturation**, I_{DS} is almost independent of V_{DS}, making the MOSFET suitable for amplification.
- **Transconductance** g_m **determines MOSFET gain**, with different expressions in linear and saturation regions.

9 The CMOS inverter DC Characteristics

- A CMOS inverter is shown below:

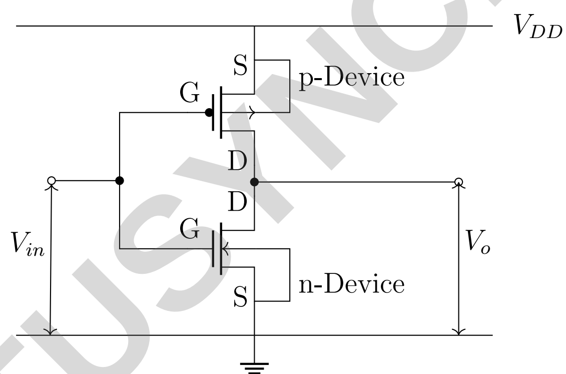

Figure 9: A CMOS inverter (with substrate connections)

- The threshold voltage of n - channel device is V_{tn}.
- The threshold voltage of p - channel device is V_{tp}. V_{tp} is negative.
- As the source of the nMOS transistor is grounded,

$$V_{gsn} = V_{in} \qquad \text{and} \qquad V_{dsn} = V_{out}$$

- As the source of the pMOS transistor is tied to V_{DD},

$$V_{gsp} = V_{in} - V_{DD} \qquad \text{and} \qquad V_{dsp} = V_{out} - V_{DD}$$

- The various regions of operations for the n - and p - transistors are shown in table below.
- For simplicity, assume $V_{tp} = -V_{tn}$
  and also assume that the pMOS transistor is 2–3 times as wide as the nMOS transistor, so $\beta_n = \beta_p$

Table 1: Relationships between voltages for the three regions of operation of a CMOS inverter

	Cutoff	Linear	Saturation
nMOS	$V_{gsn} < V_{tn}$ $V_{in} < V_{tn}$	$V_{gsn} > V_{tn}$ $V_{in} > V_{tn}$ $V_{dsn} < V_{gsn} - V_{tn}$ $V_{out} < V_{in} - V_{tn}$	$V_{gsn} > V_{tn}$ $V_{in} > V_{tn}$ $V_{dsn} > V_{gsn} - V_{tn}$ $V_{out} > V_{in} - V_{tn}$
pMOS	$V_{gsp} > V_{tp}$ $V_{in} > V_{tp} + V_{DD}$	$V_{gsp} < V_{tp}$ $V_{in} < V_{tp} + V_{DD}$ $V_{dsp} > V_{gsp} - V_{tp}$ $V_{out} > V_{in} - V_{tp}$	$V_{gsp} < V_{tp}$ $V_{in} < V_{tp} + V_{DD}$ $V_{dsp} < V_{gsp} - V_{tp}$ $V_{out} < V_{in} - V_{tp}$

- The characteristics of I_{dsn} and I_{dsp} in terms of V_{dsn} and V_{dsp} for various values of V_{gsn} and V_{gsp} can be drawn as:

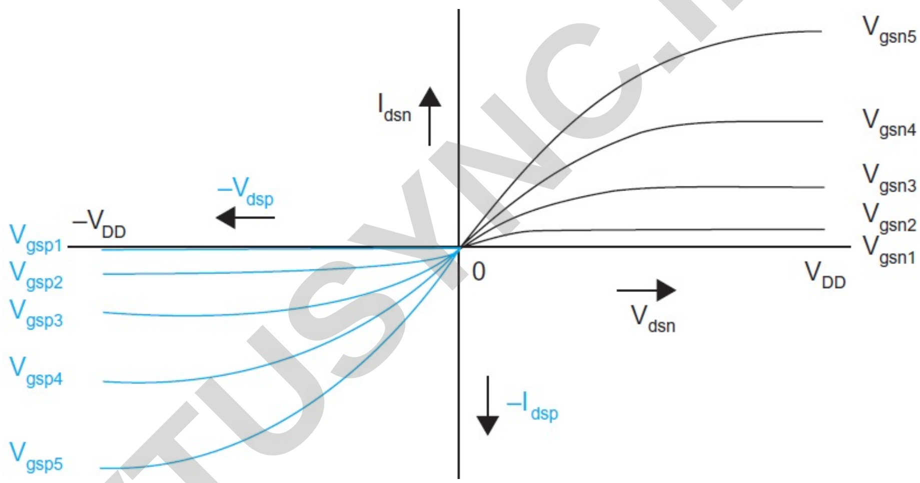

Figure 10: I - V characteristics of CMOS inverter

- Figure below shows the same plot of I_{dsn} and $|I_{dsp}|$ now in terms of V_{out} for various values of V_{in}.

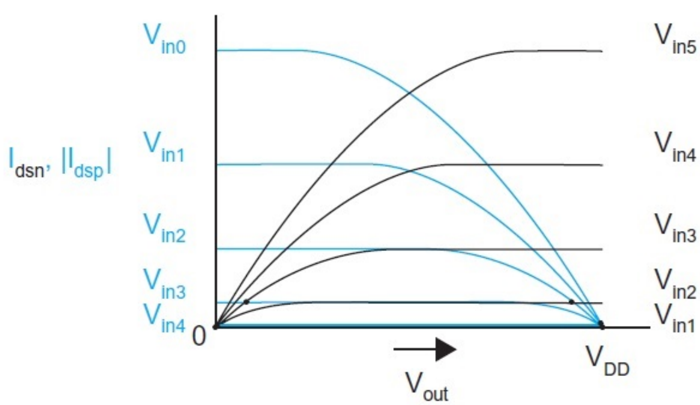

Figure 11: I_{dsn} and $|I_{dsp}|$ in terms of V_{out} for various values of V_{in}

- The possible operating points of the inverter are the values of V_{out} where $I_{dsn} = |I_{dsp}|$ for a given value of V_{in}.
- These operating points are plotted on V_{out} vs. V_{in} axes to show the inverter DC transfer characteristics.

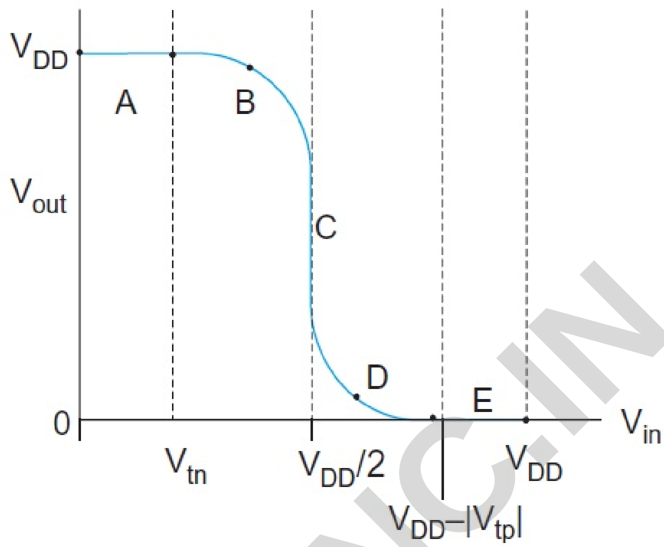

Figure 12: DC transfer characteristics of CMOS inverter

- The operation of the CMOS inverter can be divided into five regions A, B, C, D and E. The state of each transistor in each region is shown in table below.

Table 2: Summary of CMOS inverter operation

Region	Condition	p-device	n-device	Output
A	$0 \leq V_{in} < V_{tn}$	Linear	Cutoff	$V_{out} = V_{DD}$
B	$V_{tn} \leq V_{in} < V_{DD}/2$	Linear	Saturated	$V_{out} > V_{DD}/2$
C	$V_{in} = V_{DD}/2$	Saturated	Saturated	V_{out} drops sharply
D	$V_{DD}/2 < V_{in} \leq V_{DD} - \lvert V_{tp}\rvert$	Saturated	Linear	$V_{out} < V_{DD}/2$
E	$V_{in} > V_{DD} - \lvert V_{tp}\rvert$	Cutoff	Linear	$V_{out} = 0$

Regions of Operation

Region A: $V_{in} < V_{tn}$

- **NMOS**: Cut-off ($I_{dsn} = 0$)
- **PMOS**: Linear region
- **Output**: $V_o = V_{DD}$ The PMOS is fully on, pulling the output to V_{DD}, while the NMOS is off.

Region B: $V_{tn} < V_{in} < \frac{V_{DD}}{2}$

- **NMOS**: Saturation
- **PMOS**: Linear region
- **Output**: V_o decreases as V_{in} increases. The NMOS starts to conduct, and the PMOS is still in the linear region, causing the output voltage to drop gradually.

  The saturation current for the NMOS is given by:

$$I_{dsn} = \frac{1}{2}\beta_n(V_{in} - V_{tn})^2$$

  The output voltage in this region can be expressed as:

$$V_o = V_{DD} - \frac{\beta_n}{\beta_p}(V_{in} - V_{tn})^2$$

Region C: $V_{in} \approx \frac{V_{DD}}{2}$

- **NMOS**: Saturation
- **PMOS**: Saturation
- **Output**: High gain region. Both transistors are in saturation, leading to a steep transition where a small change in V_{in} results in a large change in V_o. This region is critical for the inverter's switching behavior.

  The saturation currents for the NMOS and PMOS are given by:

$$I_{dsn} = \frac{1}{2}\beta_n(V_{in} - V_{tn})^2$$

$$I_{dsp} = \frac{1}{2}\beta_p(V_{in} - V_{DD} - V_{tp})^2$$

  At the switching point ($V_{in} = \frac{V_{DD}}{2}$), the output voltage is:

$$V_o = \frac{V_{DD}}{2}$$

Region D: $\frac{V_{DD}}{2} < V_{in} < V_{DD} - |V_{tp}|$

- **NMOS**: Linear region
- **PMOS**: Saturation
- **Output**: V_o continues to decrease as V_{in} increases. The NMOS is now in the linear region, and the PMOS is in saturation, pulling the output closer to GND.

  The output voltage in this region is given by:

$$V_o = \frac{\beta_p}{\beta_n}(V_{in} - V_{DD} - V_{tp})^2$$

Region E: $V_{in} > V_{DD} - |V_{tp}|$

- **NMOS**: Linear region
- **PMOS**: Cut-off ($I_{dsp} = 0$)
- **Output**: $V_o = 0$ The NMOS is fully on, pulling the output to GND, while the PMOS is off.

Key Points

- **Switching Point**: The switching point of the CMOS inverter is typically designed to be at $\frac{V_{DD}}{2}$. This ensures a balanced transition between high and low states, maximizing noise margins.
- **Current Spike**: During the transition region (Region C), both transistors are momentarily on, leading to a short current spike from the power supply. This is known as **short-circuit current** and is a consideration in power dissipation.
- **Steep Transition**: The CMOS inverter exhibits a very steep transition between high and low states. This is desirable for digital circuits as it provides good noise immunity and a clear distinction between logic levels.
- **Noise Margins**: The steep transition also contributes to high noise margins, making CMOS inverters robust against noise and variations in supply voltage.

10 Influence of β_n/β_p ratio on transfer characteristics

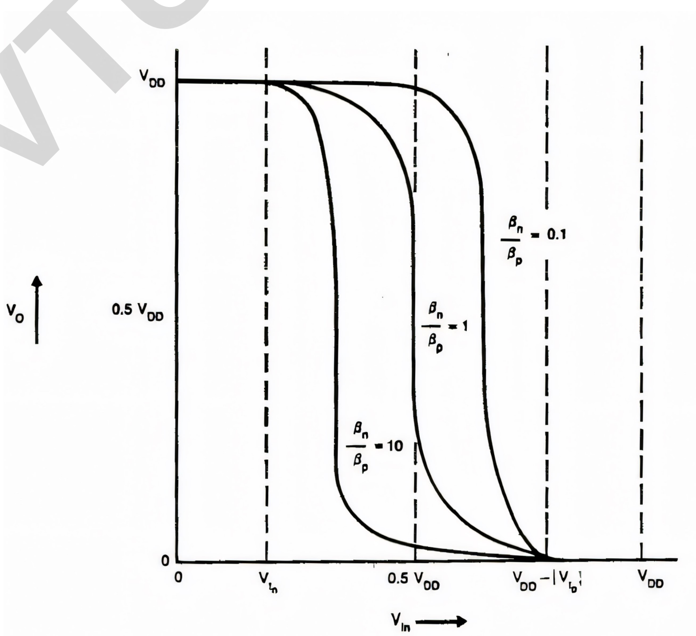

Figure 13: Influence of β_n/β_p on inverter DC transfer characteristic

- The voltage transfer characteristic (VTC) of a CMOS inverter depends on the ratio of the electron to hole mobility factors, β_n/β_p.
- The gate threshold voltage V_{inv} is defined by the condition:

$$V_{\text{in}} = V_{\text{out}} \tag{20}$$

- This threshold voltage varies with β_n/β_p.
- The transfer characteristic shifts as this ratio changes.
- Specifically, as β_n/β_p decreases, the transition region moves from left to right.
- However, the output voltage transition remains sharp, ensuring that the switching performance is not affected.
- This behavior contrasts with an nMOS inverter, where the transition gain strongly depends on the ratio of the pull-up (load) and pull-down (driver) transistors.

10.1 Adjusting β_n/β_p Through Channel Dimensions

- For a given semiconductor fabrication process, β_n/β_p can be modified by adjusting the channel dimensions—specifically, the channel length L and width W of the MOS transistors.
- A balanced design, where:

$$\frac{\beta_n}{\beta_p} = 1 \tag{21}$$

  is often desirable.
- This ensures that a capacitive load charges and discharges in equal time intervals, providing equal current source and sink capabilities.

10.2 Temperature Dependence of Transfer Characteristics

- Temperature variations influence the effective carrier mobility in MOS transistors, which in turn affects β.
- As temperature increases, carrier mobility decreases, leading to a reduction in β.
- The dependence of β on temperature T can be expressed as:

$$\beta \propto T^{-3/2} \tag{22}$$

  Therefore

$$I_{ds} \propto T^{-3/2} \tag{23}$$

- Since both electron and hole mobilities are affected similarly by temperature, the ratio β_n/β_p remains approximately constant.
- However, the threshold voltages (V_{tn} and V_{tp}) decrease with temperature.

- This causes the extent of region A in the transfer characteristic to reduce, while region E expands.
- Consequently, the overall transfer curve shifts to the left as temperature increases.
- For instance, if the temperature rises by 50°C, the threshold voltages decrease by approximately 200 mV each, leading to a 0.4 V shift in the input threshold.

10.3 Conclusion

- The β_n/β_p ratio influences the position of the transition region in the transfer characteristic of a CMOS inverter but does not degrade switching performance.
- Temperature variations cause a leftward shift in the VTC due to the reduction in threshold voltages, though the β_n/β_p ratio itself remains relatively stable.

11 Noise Margin

- Noise margin allows us to determine the allowable noise voltage on the input of a gate so that the output will not be corrupted.

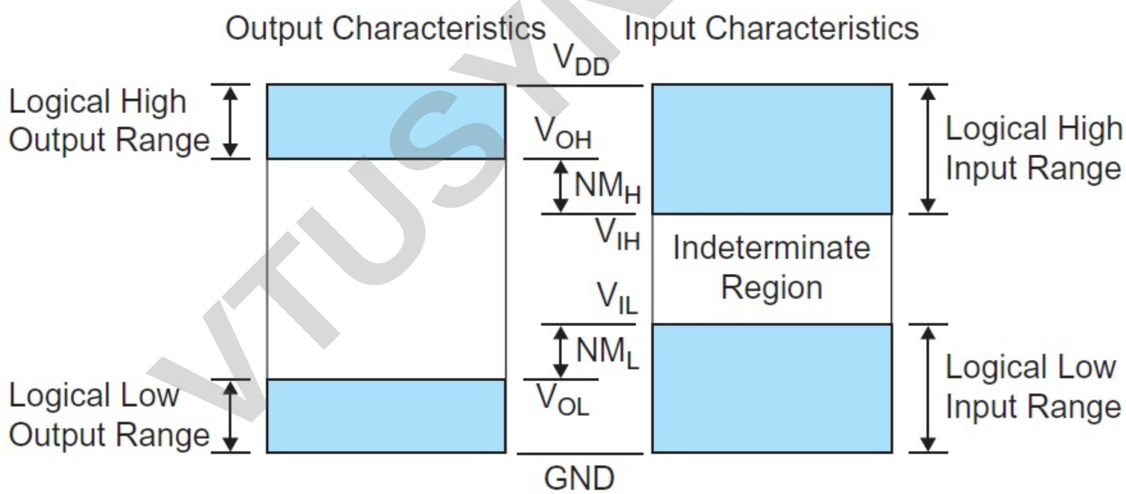

Figure 14: Noise margin definitions

- The LOW noise margin, NM_L, is defined as the difference in maximum LOW input voltage recognized by the receiving gate and the maximum LOW output voltage produced by the driving gate.

$$NM_L = V_{IL} - V_{OL} \tag{24}$$

- The high noise margin, NM_H, is the difference between the minimum HIGH output voltage of the driving gate and the minimum HIGH input voltage recognized by the receiving gate.

$$NM_H = V_{OH} - V_{IH} \tag{25}$$

Where-

V_{IH} = minimum HIGH input voltage
V_{IL} = maximum LOW input voltage
V_{OH} = minimum HIGH output voltage
V_{OL} = maximum LOW output voltage

- For an ideal inverter:

$$V_{OH} = V_{DD}$$
$$V_{IH} = \frac{V_{DD}}{2}$$
$$V_{OL} = GND$$
$$V_{IL} = \frac{V_{DD}}{2}$$

Hence for an ideal inverter:

$$NM_L = NM_H \equiv \frac{V_{DD}}{2} \tag{26}$$

12 Alternate CMOS Inverters

12.1 Pseudo-nMOS Inverter

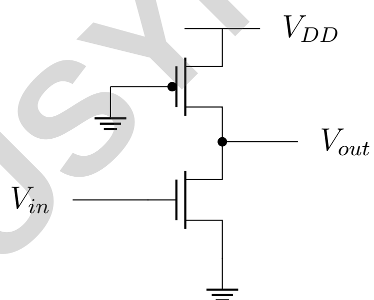

Figure 15: Pseudo-nMOS Inverter

- A **pseudo-nMOS inverter** is a variation of the standard CMOS inverter, where the **pMOS pull-up transistor has its gate permanently grounded**.
- This design is similar to using a **depletion-mode transistor** as a load in nMOS logic.

12.1.1 Characteristics

- Unlike a complementary CMOS inverter, this circuit **dissipates DC power** whenever the **nMOS pull-down transistor is turned on**.
- The **transfer characteristic** of the inverter depends on the **ratio** β_n/β_p, affecting both the **output low voltage** (V_{OL}) and the **switching threshold**.

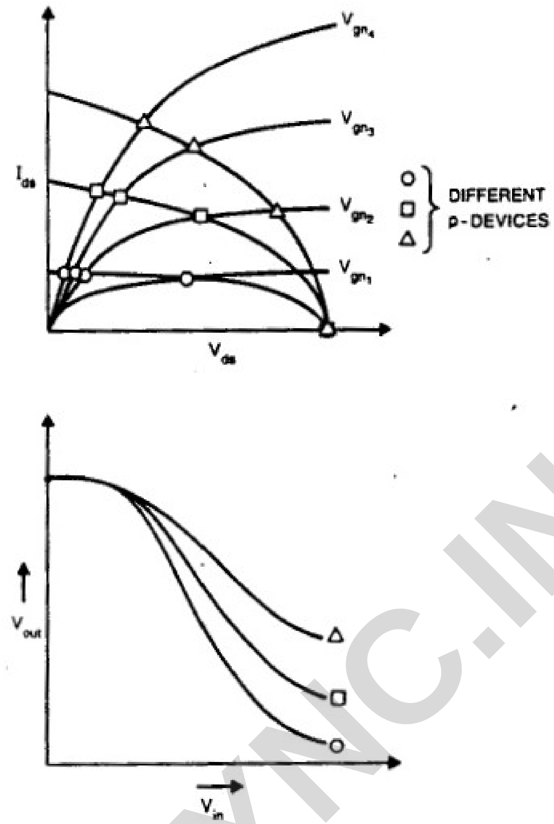

Figure 16: DC transfer characteristics of Pseudo-nMOS Inverter

12.2 Cascaded Pseudo-nMOS Inverters

- To analyze signal integrity in **pseudo-nMOS inverters**, consider two such inverters connected in series.
- Cascading inverters without signal degradation requires that the **output of one inverter should be a valid input level for the next inverter**.
- This leads to the condition:

$$V_O = V_{in} = V_{inv}$$

  where V_{inv} is the **inverter threshold voltage**.

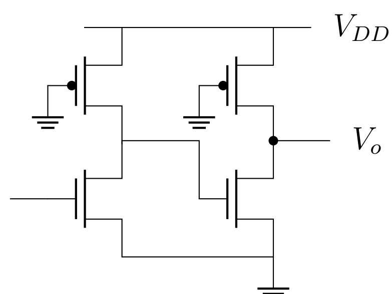

Figure 17: Cascaded Pseudo-nMOS Inverter

12.2.1 Inverter Threshold Voltage (V_{inv})

For equal noise margins, the threshold voltage should be set at approximately:

$$V_{inv} \approx 0.5V_{DD}$$

At this operating point:

- The **nMOS transistor (pull-down)** operates in **saturation**

$$0 < V_{inv} - V_{tn} < V_{dsn}$$

- The **pMOS transistor (pull-up)** operates in **linear mode**

$$0 < V_{dsp} < -V_{tp}$$

12.2.2 β-Ratio Calculation

From the MOSFET current equations:

- For the **nMOS transistor in saturation mode**:

$$I_{dsn} = \frac{1}{2}\beta_n(V_{inv} - V_{tn})^2$$

- For the **pMOS transistor in linear mode**:

$$I_{dsp} = \beta_p(V_{DD} - V_{inv})V_{inv}$$

- Equating the currents:

$$\frac{1}{2}\beta_n(V_{inv} - V_{tn})^2 = \beta_p(V_{DD} - V_{inv})V_{inv}$$

- Solving for the β-ratio:

$$\frac{\beta_n}{\beta_p} = \frac{2V_{inv}(V_{DD} - V_{inv})}{(V_{inv} - V_{tn})^2}$$

- For $V_{DD} = 5V$ and typical values $V_{inv} = 0.5V_{DD}$, $V_{tn} = 0.2V_{DD}$, we obtain:

$$\frac{\beta_n}{\beta_p} \approx 3$$

- A $4:1$ ratio can be used for slightly lower noise immunity, following a common nMOS design rule.

12.2.3 Effect of Cascading

- If the β_n/β_p ratio is too small, signal levels degrade, causing unreliable logic transitions.
- A higher ratio ensures **faster switching** and **better signal integrity** across multiple stages.
- Proper β-ratio selection allows cascaded inverters to **maintain sharp logic transitions** without excessive power dissipation.

12.2.4 Applications

Cascaded pseudo-nMOS inverters are used in:

- Precharge logic circuits
- Memory cell designs (e.g., ROMs, PLAs)
- High-density nMOS-dominated logic circuits

12.3 CMOS Tri-State Inverter

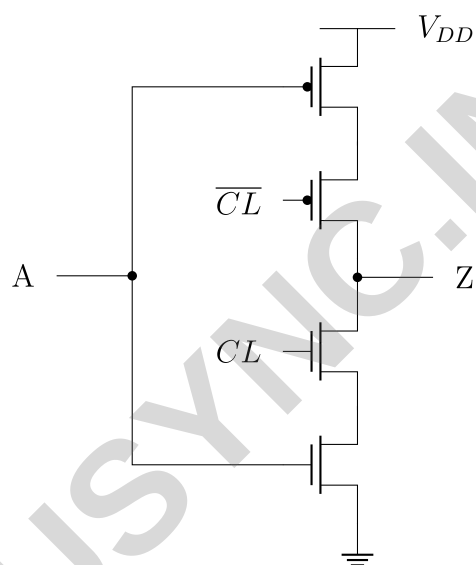

Figure 18: CMOS Tri - State Inverter

A **tri-state inverter** introduces a control mechanism, allowing the output to be either **active (logic high/low)** or in a **high-impedance (Z) state**.

12.3.1 Functionality

- When $CL = 0$, the **output is in a high-impedance (Z) state** (not driven by the input A).
- When $CL = 1$, the **output follows the input** A (acting as a regular inverter).

12.3.2 Performance Considerations

For equal-sized **nMOS and pMOS transistors**, the **tri-state inverter is approximately half the speed** of a standard CMOS inverter.

12.3.3 Applications

The tri-state inverter is widely used in:

- Clocked logic circuits

- Latches and multiplexers
- I/O structures

12.4 Summary

Inverter Type	Advantages	Disadvantages	Common Applications
Pseudo-nMOS Inverter	High transistor density, simple design	Dissipates static power	Static ROMs, PLAs
Cascaded Pseudo-nMOS Inverters	Improved logic transitions, compact design	Requires careful β-ratio selection	Precharge logic, Memory cells
CMOS Tri-State Inverter	Enables bus sharing, supports clocked logic	Slower than standard CMOS inverter	Latches, Multiplexers, I/O structures

These inverters are essential in **low-power, high-density digital circuits**, where trade-offs between **power consumption, speed, and circuit complexity** must be carefully considered.

13 Transmission gate DC characteristics

- A transmission gate is a complementary switch that consists of an **n-channel MOS (nMOS)** transistor and a **p-channel MOS (pMOS)** transistor connected in parallel.
- Each transistor has separate gate connections but shares common source and drain terminals.
- The control signal ϕ is applied to the **gate of the nMOS transistor**, while its complement $\bar{\phi}$ is applied to the **gate of the pMOS transistor**.

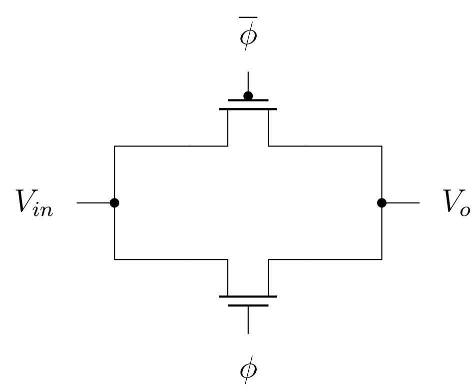

Figure 19: Transistor connection for CMOS transmission gate

13.1 nMOS Pass Transistor Characteristics

The behavior of an **nMOS transistor** as a pass transistor can be analyzed in two cases:

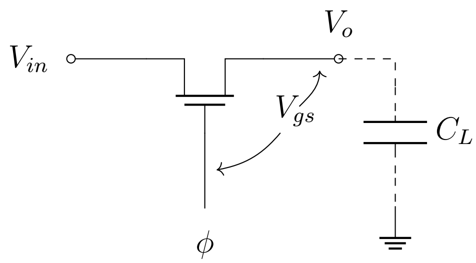

Figure 20: nMOS Transistor in transmission gate

Case I: Transmission of Logic '1' ($V_{in} = V_{DD}$)

- Initially, the load capacitor C_L is discharged ($V_O = 0$).
- When $V_{in} = 1$ and $\phi = 1$, the nMOS transistor turns **on** and starts charging the capacitor.
- Current flows until V_O approaches $V_{DD} - V_t$, where V_t is the **threshold voltage**.
- Since V_O is limited to $V_{DD} - V_t$, the transmission of logic '1' is **degraded**.

Case II: Transmission of Logic '0' ($V_{in} = 0$)

- When $V_{in} = 0$ and $\phi = 1$, the nMOS transistor discharges the capacitor.
- As V_O approaches 0V, the current diminishes.
- **Logic '0' is transmitted accurately.**

13.2 pMOS Pass Transistor Characteristics

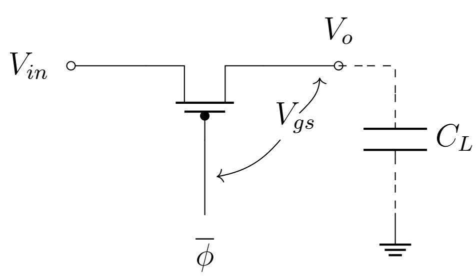

Figure 21: pMOS Transistor in transmission gate

Similarly, for a **pMOS transistor**, the behavior of a **pMOS transistor** as a pass transistor can be analyzed in two cases:

Case I: Transmission of Logic '1' ($V_{in} = V_{DD}$)

- Initially, the load capacitor C_L is discharged ($V_O = 0$).

- When $V_{in} = 1$ and $\bar{\phi} = 0$, the pMOS transistor turns **on** and starts charging the capacitor.
- Current flows until V_O approaches V_{DD}, meaning the transmission of logic '1' is **strong and accurate**.

Case II: Transmission of Logic '0' ($V_{in} = 0$)

- When $V_{in} = 0$ and $\bar{\phi} = 0$, the pMOS transistor turns **on**, allowing the capacitor to discharge.
- As V_O approaches V_t, where V_t is the **threshold voltage**, the transistor ceases conduction.
- Since V_O is limited to V_t, the transmission of logic '0' is **degraded**.

Table 3: Transmission characteristics of n-channel and p-channel pass transistors

Device	Transmission of '1'	Transmission of '0'
n	poor	good
p	good	poor

13.3 Combining nMOS and pMOS — The Transmission Gate

- The transmission gate is a CMOS switch that combines an nMOS transistor and a pMOS transistor in parallel to capitalize on their complementary strengths.
- When these two transistors are combined:
  - They ensure that both logic levels are transmitted without degradation.
  - Both devices are controlled by complementary gate signals derived from a single control signal ϕ.
  - When $\phi = 1$, both transistors turn ON, allowing the input V_{in} to be accurately transferred to the output V_{out}.
  - When $\phi = 0$, both transistors are OFF, isolating the input from the output and presenting a high impedance state.
- The overall operation is shown in table below:

Table 4: Operation of a Transmission Gate

Control Signal	nMOS	pMOS	$\mathbf{V_{in}}$	$\mathbf{V_o}$
$\phi = 0, \overline{\phi} = 1$	OFF	OFF	0	Z (High Impedance)
$\phi = 0, \overline{\phi} = 1$	OFF	OFF	1	Z (High Impedance)
$\phi = 1, \overline{\phi} = 0$	ON	ON	0	0 (Strong)
$\phi = 1, \overline{\phi} = 0$	ON	ON	1	1 (Strong)

- The corresponding output characteristic is shown in figure below.

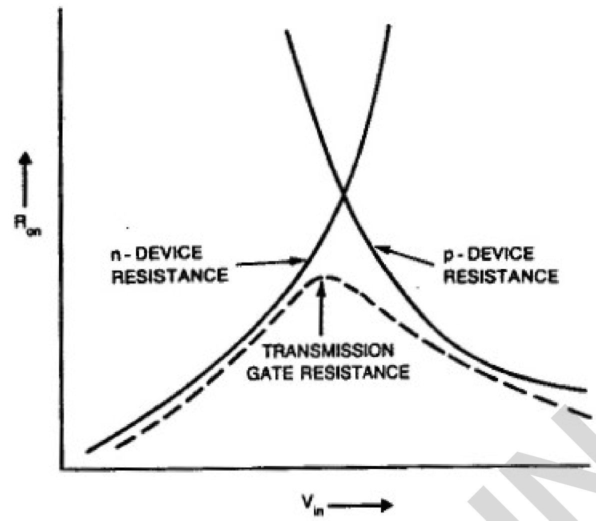

Figure 22: Transmission gate output characteristic

14 Latch-up in CMOS

- Latch-up is a parasitic circuit effect that causes a low-resistance path between power supply rails (VDD and VSS) in CMOS circuits.
- This results in excessive current flow, leading to circuit failure or permanent damage.
- Early CMOS processes were highly susceptible to latch-up, but modern fabrication techniques and circuit design strategies have significantly mitigated this issue.

14.1 Mechanism of Latch-up

- Latch-up occurs due to the presence of **parasitic bipolar junction transistors (BJTs)** within the CMOS structure.
- These unintended transistors can form a positive feedback loop, causing a short circuit between VDD and VSS.

14.1.1 Parasitic Bipolar Transistor Formation

A CMOS structure consists of nMOS transistors in a *p-well* and pMOS transistors in an *n-substrate*. Two parasitic bipolar transistors—**npn** and **pnp**—are inadvertently formed:

- The **npn transistor** consists of:
  - **Emitter**: n+ source/drain of nMOS
  - **Base**: p-well
  - **Collector**: n-substrate

- The **pnp transistor** consists of:
  - **Emitter**: p+ source/drain of pMOS
  - **Base**: n-substrate
  - **Collector**: p-well

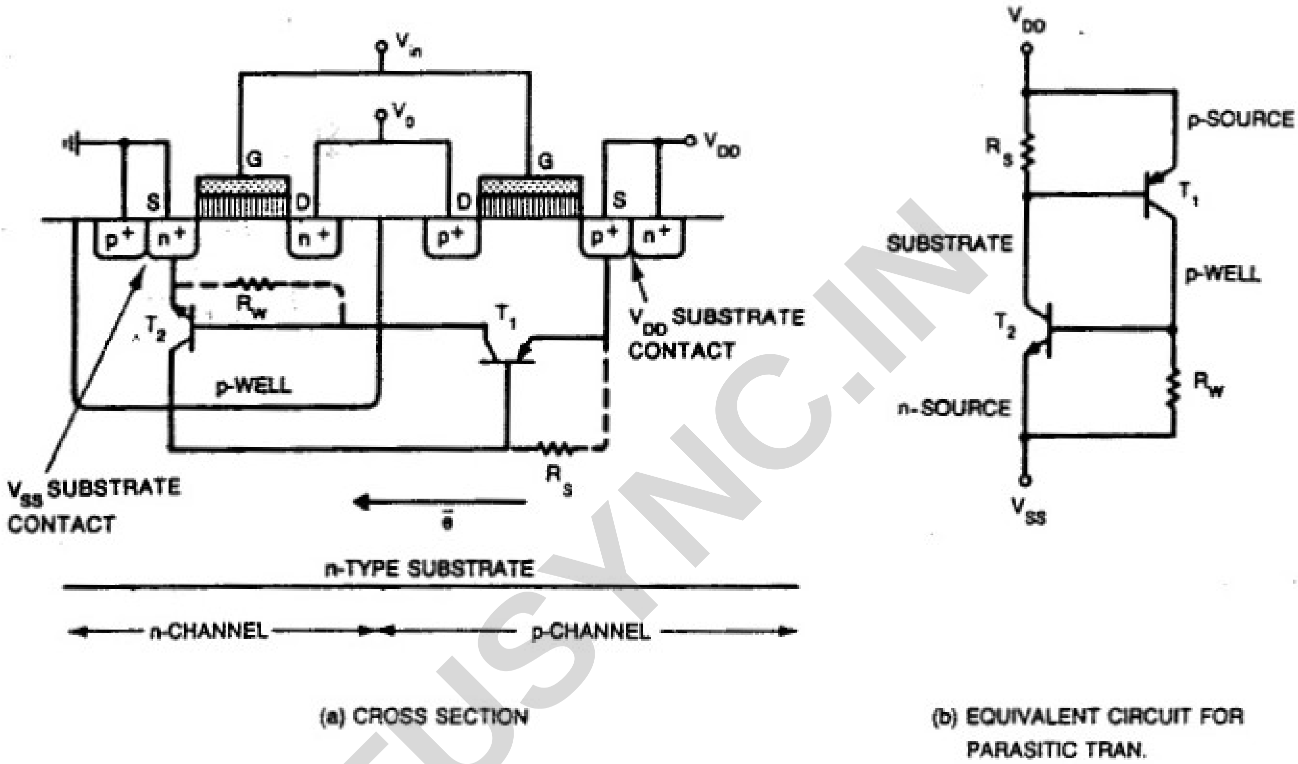

Figure 23: Source of latch-up In CMOS

14.1.2 Latch-up Condition

Latch-up occurs when:

1. **Excess carrier injection**: Voltage spikes or transient currents inject minority carriers into the substrate or well.

2. **Voltage drop across Rs and Rw**: If sufficient voltage develops, it can forward bias the base-emitter junctions of the parasitic transistors.

3. **Positive feedback activation**: The feedback loop between the npn and pnp transistors sustains the low-resistance state.

14.2 Preventive Measures

To prevent latch-up, the following design strategies are implemented:

14.2.1 Substrate and Well Contacting

- Every well should have a **substrate contact**.
- Substrate contacts should be directly connected to supply pads using metal.
- Contacts should be placed **close to transistor source connections** to reduce resistance.

14.2.2 Transistor Placement Guidelines

- Maintain **separation** between nMOS and pMOS devices.
- Group nMOS transistors closer to **VSS** and pMOS transistors closer to **VDD**.
- Avoid **checkerboard-style layouts**.

14.2.3 I/O Structure Design

- Use **guard rings** (p+ around nMOS, n+ around pMOS).
- Employ **minimum-area p-wells** to reduce photocurrent injection.
- Hard-wire the **p-well to ground (via p+ contact)**.
- Reduce **spacing between p-well and nMOS source contact**.

14.3 Conclusion

- Modern CMOS processes incorporate design strategies such as proper substrate contacting, optimized transistor placement, and structured I/O design to ensure latch-up immunity.
- By following these best practices, latch-up failures can be minimized.